Healthcare Appointment No-Show Prediction

Project Report

1. Introduction

Missed healthcare appointments can waste clinical resources, delay care, and make scheduling less efficient. This project explores appointment data to understand no-shows and demonstrate how data analytics and machine learning may support better appointment planning. The project combines a Python-based analysis and prediction model with a Power BI dashboard for presenting key findings.

2. Abstract

A hospital appointment dataset was analysed to examine appointment outcomes and build an initial model to classify no-show appointments. Records labelled “No-show” and “Completed” were selected for the modelling step; cancelled and scheduled appointments were not included in that binary target. After preparing the data, a Decision Tree classifier was trained and evaluated. The initial test accuracy was 60%, indicating that the model has room for improvement and should not be used on its own for operational decisions. A Power BI dashboard was then created to communicate appointment status, monthly and weekday no-shows, and a comparison of actual and predicted outcomes.

3. Tools Used

- **Python:** data preparation and analysis.
- **Google Colab:** cloud notebook environment for running Python.
- **Pandas:** loading, cleaning, and transforming appointment data.
- **Scikit-learn:** training and evaluating the Decision Tree model.
- **Power BI:** building interactive-style visuals and summarising results.

4. Steps Involved in Building the Project

1. **Data import:** Loaded the hospital appointments CSV file into Google Colab using Pandas.
2. **Data inspection and cleaning:** Checked the dataset structure, missing values, duplicate appointment IDs, and date fields. Converted appointment dates to a date format for analysis.
3. **Feature preparation:** Used available fields to create modelling features, including doctor, reason for visit, appointment month, weekday, and appointment hour. Kept completed and no-show records for the binary prediction task.
4. **Model training and evaluation:** Split the prepared data into training and test sets, trained a Decision Tree classifier, and assessed its predictions. The initial test accuracy was 60%, so further data and model improvements would be needed.
5. **Power BI dashboard:** Imported the analysis and prediction CSV files and created visuals for appointment status, monthly no-shows, weekday patterns, actual versus predicted results, and prediction correctness.
6. **Interpretation:** Used the dashboard to present patterns clearly and identify areas for further investigation, such as reminder practices, appointment timing, and additional patient or scheduling information where appropriately available.

5. Conclusion

This project provided practical experience in preparing healthcare data, building and evaluating a basic machine learning model, and communicating results through Power BI. The dashboard makes appointment patterns easier to review, while the initial model demonstrates the prediction workflow. Since the model achieved 60% test accuracy on a small selected dataset, its predictions are exploratory rather than reliable clinical or scheduling advice. Future work could include a larger and more representative dataset, additional relevant features, comparison with other models, and validation before any real-world use. Overall, the project demonstrates how healthcare data analytics can help inform service improvement.